

CUSTOMER FUTURE CLV PREDICTION

Viva Questions and Answers

Project-specific viva preparation based on the completed notebook workflow and recorded results.

1. What is the objective of your project?

Answer: To predict Future Customer Lifetime Value using historical customer purchasing behaviour and then segment customers according to predicted value.

2. What type of ML problem is this?

Answer: It is a supervised machine learning regression problem because Future_CLV is a continuous numerical target.

3. What is CLV?

Answer: Customer Lifetime Value is the estimated value a customer contributes to a business over time.

4. What is the target variable?

Answer: Future_CLV.

5. How did you create the target?

Answer: I calculated Amount as $\text{Quantity} \times \text{Price}$, grouped future-period transactions by Customer ID, summed Amount, and renamed the result Future_CLV.

6. What is RFM?

Answer: RFM stands for Recency, Frequency, and Monetary value.

7. What is Recency?

Answer: The number of days between the snapshot date and the customer's most recent purchase.

8. What is Frequency?

Answer: The number of purchase events or transactions made by a customer.

9. What is Monetary?

Answer: The total historical spending or value generated by a customer.

10. What additional features did you use?

Answer: Average_Order_Value, Total_Quantity, Unique_Products, Active_Months, and Avg_Purchase_Interval.

11. How many customer records were in the feature table?

Answer: 3,313 customer records.

12. How many customers were in the target dataset?

Answer: 2,868 customers.

13. What did the target distribution show?

Answer: A highly right-skewed distribution with a long right tail caused by a small number of extremely high-value customers.

14. What is a correlation heatmap?

Answer: A visualization of correlation coefficients showing positive and negative relationships between variables.

15. Which feature correlated most strongly with Future_CLV?

Answer: Monetary, with an observed correlation of approximately 0.80.

16. What was the correlation of Frequency with Future_CLV?

Answer: Approximately 0.60.

17. What was the relationship between Recency and Future_CLV?

Answer: Negative, approximately -0.15.

18. Why was Average_Order_Value weakly correlated with Future_CLV?

Answer: Average order value alone does not capture purchase frequency or total spending.

19. What is an outlier?

Answer: A value that is unusually far from the majority of observations.

20. How did you detect outliers?

Answer: Using boxplots and the Interquartile Range method.

21. What is IQR?

Answer: IQR equals Q3 minus Q1.

22. What are IQR outlier bounds?

Answer: Lower bound = $Q1 - 1.5 \times IQR$ and upper bound = $Q3 + 1.5 \times IQR$.

23. Why did you not remove all outliers?

Answer: Extreme customers may be genuine and commercially important high-value customers.

24. Which feature had zero detected IQR outliers?

Answer: Recency.

25. What is a baseline model?

Answer: A simple benchmark used to determine whether ML models provide meaningful improvement.

26. Which baselines did you use?

Answer: Mean baseline and median baseline.

27. Mean baseline results?

Answer: MAE 955.96 and RMSE 3915.39.

28. Median baseline results?

Answer: MAE 820.43 and RMSE 3964.10.

29. Why did the median baseline have lower MAE?

Answer: The median is less influenced by extreme values in the skewed target.

30. What is Random Forest?

Answer: An ensemble algorithm that combines predictions from many decision trees.

31. Why use Random Forest?

Answer: It handles non-linear relationships and feature interactions and provides feature importance.

32. Random Forest baseline results?

Answer: MAE 644.12 and RMSE 2639.68.

33. Random Forest with log features results?

Answer: MAE 641.85, RMSE 2609.34, and R^2 0.5552.

34. Why experiment with log features?

Answer: To reduce the influence of strongly right-skewed values and compress large numerical ranges.

35. What is XGBoost?

Answer: A gradient boosting algorithm that builds trees sequentially to correct earlier prediction errors.

36. Why use XGBoost?

Answer: It is powerful for structured tabular data and captures complex non-linear relationships.

37. XGBoost baseline results?

Answer: MAE 663.78, RMSE 2282.51, and R^2 0.6596.

38. Tuned XGBoost results?

Answer: MAE 616.57, RMSE 2640.58, and R^2 0.5445.

39. Why did you select baseline XGBoost?

Answer: It had the best recorded RMSE and R^2 , giving the best overall performance despite tuned XGBoost having lower MAE.

40. What is MAE?

Answer: The average absolute difference between actual and predicted values.

41. What is RMSE?

Answer: The square root of average squared prediction errors; it penalizes large errors more heavily.

42. What is R^2 ?

Answer: A measure of how much variation in the target is explained by the model.

43. What does $R^2 = 0.6596$ mean?

Answer: The model explained about 65.96% of the observed target variation in the evaluated experiment.

44. Why is RMSE important here?

Answer: Because extreme CLV prediction errors are commercially important and RMSE penalizes large errors strongly.

45. What is feature importance?

Answer: A measure of how strongly features contribute to model predictions.

46. Most important Random Forest feature?

Answer: Monetary.

47. Most important XGBoost features?

Answer: Monetary followed by Total_Quantity.

48. What does an Actual vs Predicted plot show?

Answer: It compares true values with predictions; ideal points lie close to the $y=x$ diagonal.

49. What is a residual?

Answer: Actual value minus predicted value.

50. What should a good residual plot look like?

Answer: Residuals should be randomly scattered around zero without strong patterns.

51. What did residual analysis reveal?

Answer: The largest errors occurred mainly for extreme high-value customers.

52. Why are extreme customers difficult to predict?

Answer: They are rare and the model has fewer similar examples from which to learn.

53. What is overfitting?

Answer: Learning training-specific noise or patterns so closely that performance on unseen data becomes worse.

54. Does hyperparameter tuning always improve a model?

Answer: No. In this project tuned XGBoost improved MAE but worsened RMSE and R^2 .

55. Difference between Random Forest and XGBoost?

Answer: Random Forest builds trees independently; XGBoost builds trees sequentially to correct previous errors.

56. Did tree models require scaling?

Answer: Generally no, because tree splits use thresholds rather than distance calculations.

57. Why can predictions be negative?

Answer: Regression models are not automatically constrained to positive values; production CLV predictions can be clipped to zero.

58. Why were some predictions identical?

Answer: Tree-based models can assign identical values to customers falling into similar terminal regions.

59. How did you segment customers?

Answer: Into Low Value, Medium Value, and High Value groups based on predicted Future CLV.

60. Why is segmentation useful?

Answer: It enables different marketing and retention strategies for customers with different expected value.

61. Strategy for high-value customers?

Answer: Personalized retention, premium service, loyalty benefits, and proactive support.

62. Strategy for medium-value customers?

Answer: Cross-selling, upselling, repeat-purchase incentives, and loyalty programs.

63. Strategy for low-value customers?

Answer: Low-cost automated engagement and selective reactivation campaigns.

64. Why were segments approximately equal in size?

Answer: The chart suggests rank-based or quantile-based segmentation.

65. What is data leakage?

Answer: Using future or target-period information during training, causing unrealistically good performance.

66. How can leakage be avoided?

Answer: Use only information available before the prediction cutoff and create the target from a future period.

67. Why is time-based splitting important?

Answer: It simulates the real-world task of predicting the future using only past information.

68. What are the main limitations?

Answer: Skewed target distribution, rare extreme customers, limited external features, and prediction uncertainty.

69. How can the project be improved?

Answer: Time-aware validation, target transformation, LightGBM/CatBoost, more features, SHAP, and business-specific segment thresholds.

70. What is SHAP?

Answer: A model explainability method that estimates feature contributions to individual predictions.

71. What is the main business insight?

Answer: Historical spending, total quantity purchased, and frequency are the strongest predictors of future customer value.

72. What is your final model?

Answer: Baseline XGBoost.

73. Give a short project summary.

Answer: I created customer-level RFM and behavioural features, predicted Future CLV using regression models, compared baselines, Random Forest, and XGBoost, selected baseline XGBoost based on RMSE and R^2 , analysed predictions and feature importance, and segmented customers by predicted future value.